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COVID-19 Disruption Gets Inside the Family: A Two-Month Multilevel Study of Family Stress During the Pandemic

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Developmental research during COVID-19 suggests that pandemic-related disruptions in family relationships are associated with children's mental health. Most of this research has focused on 1 child per family, thereby obfuscating patterns that are differentially operative at the family-wide (i.e., between-family) versus child-specific (i.e., within-family) levels of analysis. Thus, the current study evaluates multilevel, longitudinal associations between COVID-19 disruption, family relationships, and caregiver/child mental health using a sibling comparison methodology. Caregivers (N = 549 families with 1098 children between 5 and 18 years old) were recruited from the Prolific research panel (73% White-European; 68% female; 76% United Kingdom, 19% U.S.A.; median 2019 income $50,000–$74,999). Caregiver reports of COVID-19 disruption, psychological distress, family functioning, parenting, and child mental health (for 2 children per family) were provided during May (time 1) and July (time 2) 2020. A Bayesian multilevel path analysis with random effects revealed: (a) families were experiencing difficulties across domains when COVID-19 disruption was high; (b) COVID-19 disruption corresponded to greater sibling differences in mental health; and (c) the sibling with poorer mental health received lower quality parenting over time, especially in families who reported higher levels of differential parenting. Findings suggest that understanding children's mental health difficulties during COVID-19 requires a family system lens due to the multiple ways these consequences permeate across the family unit. Comprehensive interventions for children's mental health during this time will likely require an examination of caregiver, sibling, and whole-family dynamics in the context of evidence-based telehealth practice.

Keywords: COVID-19, children, family, mental health, siblings

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The global spread of SARS-CoV-2 has created unprecedented stress for individuals and families around the world (Gassman-Pines et al., 2020; Patrick et al. 2020). Households have faced economic recession, job loss, social distancing and isolation, crowding, childcare and school closures (and reopening), travel restrictions, and resource scarcity, in addition to the direct biomedical threat of the COVID-19 disease. Not surprisingly, prevalence rates for mental health problems among children and caregivers—including anxiety, depression, and posttraumatic stress symptomatology—have been high during COVID-19 relative to prepandemic estimates (Cooke et al., 2020; Patrick et al. 2020; Racine, Cooke, et al., 2020), though additional studies on prevalence change are warranted.

Despite a valiant response from the psychological community, there is a need to better understand the mechanisms through which COVID-19 disruption is related to child and family functioning, in addition to longitudinal patterns throughout development (Wade et al., 2020). Moreover, differential patterns for whole families (between-family effects) versus particular individuals (within-family effects) have not been parsed, which limits our understanding of the etiological pathways to child psychopathology (D'Onofrio et al., 2013). That is, we do not know how individual (i.e., sibling) and family differences modify associations with pandemic-related stress. In studies of one child per family, these sources of variability are not isolated, leading to potentially erroneous generalizations from children to families, and vice versa. Furthermore, most COVID-19 developmental science has been cross-sectional. Thus, the direction of association between children and caregivers has yet to be identified (Wade et al., 2020). This is important for planning public health

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policy and disaster response services, while potentially informing novel interventions to support children and families during periods of stay-at-home orders and social distancing.

To address these knowledge gaps, the present study tests a multilevel (between-family and within-family), longitudinal model of family stress during pandemic. Specific attention is paid to: (a) the associations between COVID-19 disruption and various areas of family life, including child/caregiver mental health, family functioning, and parenting quality; (b) the magnitude of sibling differences in mental health and quality of parenting behaviors directed toward them; and (c) within-family (i.e., child-specific) associations between mental health and parenting quality through the comparison of siblings.

Family Spillover and Transactional Processes in Development

Family processes are considered an important mechanism through which adversity influences family members, both individually and relationally (Walsh, 2015). This phenomenon has been described in several ways, perhaps most poignantly through the concepts of spillover in family systems and transactional processes in human development. Keeping with the systemic tenets of family systems theory (Carr, 2015; Fiese et al. 2019), *spillover effects* occur when emotional turmoil that initially influences one individual or dyad in the family subsequently influences another individual or dyad as the hardship propagates throughout the family system. These processes often manifest through the relational transmission of psychosocial stress. In one example, Grzywacz and colleagues (2002) described spillover as occurring when parents have a stressful day at work, which corresponds to lower quality family interactions in the evenings as caregivers are stressed, tired, and irritable. Spillover can also occur within the home, whereby conflict within one family subsystem (e.g., the marital unit) is related to behavior within another domain (e.g., parent–child relationships or child behavior; Jenkins et al. 2003; Sears et al. 2016). Through these interdependencies, a shared family climate of inter- and intrapersonal experience arises among family members (Hollenstein et al., 2004; Repetti et al. 2002). This emergent property of the entire family is described as something like *the whole is greater than the sum of its parts* (Minuchin & Fishman, 1981).

Complementary to the concept of family spillover is the theory of *transactional processes*, or the principle that child development occurs through dynamic and recursive interactions between children and their environments (Sameroff & Mackenzie, 2003). This process is bidirectional and reciprocal, wherein "... the child's behavior changes the caregiver's expectations and behavior and is, in turn, changed by the changed caregiver" (Sameroff & Mackenzie, 2003; p. 614). Plamondon and colleagues (2018) identified transactional effects in a longitudinal study of siblings, whereby the child who had higher levels of behavioral difficulties elicited more parental harshness, which predicted a subsequent increase in behavioral problems. Notions of spillover and transactional processes in development are important for understanding the impact of COVID-19 stress/disruption on child adjustment, necessitating the consideration of interdependencies, whole family processes, and reciprocal influences of children and parents on one another. Consistent with spillover/transactional models, the current study examines bidirectional associations between children's mental health and parenting quality, in addition to multiple indirect pathways linking COVID-19 disruption and family processes.

From the Family Stress Model to the COVID−19 Family Disruption Model

Family spillover processes have been described in a cascading fashion, from distal to proximal risk, through the Family Stress Model (Conger et al., 1994). In this framework, family economic pressure is linked to child developmental health via heightened psychological distress of caregivers and compromised family relationship quality (Masarik & Conger, 2017). During times of financial duress, caregivers have fewer psychological resources (e.g., time, attention, patience). Consequently, marital and parent–child relationships can become more hostile and less sensitive. The consequences of these economic disruptions in family relationships cut across domains of child development. Specifically, the family stress framework has been replicated across child outcomes, study settings, and diverse populations, including emotional and behavioral problems in Midwestern youth (Conger et al., 1994) and African American children (Conger et al., 2002), cognitive development and socioemotional problems in US newcomer children (Mistry et al., 2008), and substance use in Mexican American youth (Martin et al., 2019). Similar findings have been observed in representative samples in the US (Yeung et al., 2002) and Canada (Kohen et al., 2008).

As in the classic family stress model, family spillover processes are gathering increased attention in the context of COVID-19. For example, Daks and colleagues (2020) modeled a cascade (in a cross-sectional study) linking COVID-19 disruption to family functioning, parenting, and child/caregiver mental health. While important, cross-sectional studies of children during the pandemic have limited capacity to test directional associations, and cannot answer questions about development (e.g., intraindividual change). There is a pressing need for additional longitudinal studies that can speak to the directionality, timing, and persistence of effects during and after the pandemic.

The family stress model is generalizable beyond direct financial insecurity of households, as there is evidence for relationship disruptions corresponding to macrocontextual stressors including economic recessions (Schneider et al., 2017) and disasters (Masten & Motti-Stefanidi, 2020). Thus, the diffuse worry felt by caregivers (Brown et al., 2020; Russell et al. 2020) may be implicated in the family stress cascade in a fashion similar to financial threat. In reviewing the literature on natural and human-made disasters (including recessions), Prime et al. (2020) outlined a conceptual framework linking social disruption due to COVID-19 with child adjustment via compromised caregiver well-being and family processes, including general family functioning and parent–child relationship quality. The current study represents an empirical test of this COVID-19 Family Disruption Model, which instantiates a pandemic-specific extension of the classical family stress framework. Keeping with concepts of spillover and transactional effects, it was hypothesized that COVID-19 disruption will be related to elevated stress across individuals, dyads, and whole families, and that the effects will be bidirectional between children and caregivers.

Similar to other prominent historical stressors (e.g., SARS, the financial crisis of 2008; Hurricane Katrina, 9/11), the hypothesized model is informed by events and disasters that have been empirically linked to a deterioration in child/caregiver mental health and

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family functioning. For example, Schneider et al. (2017) demonstrated that there were increases in incidences of maltreatment following the 2008 global recession, putatively due to increases in caregiver stress and a deterioration in the quality of household relationships. In terms of natural disasters, Kelley and colleagues (2010) found that loss/disruption associated with Hurricane Katrina predicted parental maladaptive coping, use of corporal punishment, and child PTSD symptoms over one year later. Parenting practices may be particularly disrupted by large-scale stressors due to the demand of ensuring family survival, which may take precedence over warm, sensitive, or responsive parenting. That said, parental positivity and involvement, if preserved, are viewed as protective factors in the face of such adversity (Kessel et al., 2019). Given the role of the family system in supporting adaptation in response to disaster (Masten & Motti-Stefanidi, 2020), the current study represents an important empirical test of these longitudinal associations in the context of COVID-19.

Increased Sibling Differences in Stressful Contexts

Macrocontextual stressors (such as poverty or disaster) are operative in the distal ecology and shared across siblings within the same household (i.e., many COVID-19 stressors in families should be similar for children residing under the same roof). However, behavioral genetic research shows that siblings are more different than similar once accounting for genetic influence, and that nonshared environments account for a greater proportion of variance in virtually all developmental outcomes including mental health (Plomin, 2011). This discrepancy is partially resolved by the replicated finding that sibling differences are actually exacerbated under conditions of family stress (Jenkins et al., 2003, 2016).

Family risk factors can enhance sibling differences through potential effects on children's biology and social environments. Aizer et al. (2016) examined variation in maternal cortisol levels over different pregnancies. They found that mothers of low socioeconomic status have greater cortisol variation across pregnancies and that this variation predicted sibling differences in IQ. Socioeconomic stress also leads to consequences for the caregiving environment, in part through these biological pathways. Browne et al. (2018) found that sibling differences in birth weight (explained by socioeconomic status) were associated with differences in the observed maternal sensitivity that siblings received, with lower birth weight siblings receiving less sensitivity. Thus, family stressors can result in less optimal biological development and differential treatment of siblings. Differential treatment (aka differential parenting) of children has a cascading impact on the family, predicting poorer sibling and family relationships over the life course (Jenkins et al., 2012; Jensen & McHale, 2017), marital conflict (Jenkins et al., 2003), and physical and mental health problems in children (Browne & Jenkins, 2012; Deater-Deckard et al., 2001). Collectively, these findings lead to the hypothesis that COVID-19 disruption (at the family-level) would be associated with greater sibling variation in parenting and mental health.

The Present Study

This study explored longitudinal associations between child/parent mental health, parenting quality, and family functioning in relation to COVID-19 disruption over a two-month window of the pandemic. The COVID-19 Family Disruption Model (Prime et al., 2020) was used as a guiding framework. Of note, the model in the current study was multilevel in that both between-family (family-wide) and within-family (child-specific) associations were considered through multilevel modeling. We tested three primary hypotheses: (1) COVID-19 disruption will be associated with higher caregiver distress, family dysfunction, and child mental health problems, and lower levels of parenting quality, over time; (2) COVID-19 disruption will correspond to greater sibling differences in parenting and mental health problems, over time; and (3) within families, children with greater mental health problems will receive lower quality parenting (and vice versa), and these longitudinal child-specific associations will be most pronounced under settings of higher between-family stress. These hypotheses, and the study procedures to test them, have not been preregistered, as this study was a rapid-response endeavor conducted in the immediate aftermath of COVID-19. The setting is a cross-national (primarily US and U.K.) online and prospective longitudinal cohort of caregivers and children. Notably, while there have been international differences in COVID-19 spread, management, and political and health care responses, modeling these differences was not the goal of this study. It is possible that some of the observed associations (e.g., between pandemic-related distress and outcomes) are partially reflective of unmeasured international differences at the time of assessment. Results should be considered within the context of national variability in pandemic severity and potential between-country confounds (Hale et al., 2021).

Method

Participants and Procedure

Participants are from the Child Resilience and Managing Pandemic Emotional Distress (CRAMPED) Families Study. This is a prospective cohort of 549 caregivers (age: M = 41.33, SD = 6.33) and 1,098 children (younger child: M = 9.62, SD = 3.21, 45.9% female; older child: M = 11.80, SD = 3.32, 49.0% female), mobilized to understand child and family mental health during the COVID-19 pandemic. Respondents were recruited via Prolific, which is an online research panel company located in the U.K. For the purposes of studying between- and within-family processes, we limited recruitment to caregivers with at least two children between the ages of 5 to 18 years. Caregiver respondents were mostly married/common-law (90%), White-European (73%), females (68%), who resided in either the United Kingdom (76%), United States (19%), Canada (4%) or Australia (1%). The current study draws on data from the time 1 (T1: May 2020) and time 2 (T2: July 2020). Readers should exercise caution when generalizing findings beyond this sample, which has limitations in terms of representativeness and timing.

Following recruitment, participants completed questionnaires online via Qualtrics, which included measurement of disruptions related to COVID-19, psychological distress, family functioning, parenting, and child mental health problems. Caregivers reported on themselves and two children in their household. See Table 1 for sociodemographic characteristics. This study received ethics approval from the Human Research Ethics Committee at the University of Waterloo (Title: Child Resilience and Managing Pandemic Emotional Distress in Families [CRAMPED]; Protocol Number: 42112).

Table 1
Sociodemographic Characteristics of Caregivers and Children at Time 1 (May 2020)

	Caregivers (n = 549)		Children (n = 1,098)	
Sociodemographic characteristic	*n*	*M (SD)*	*n*	*M (SD)*
Age	527	41.33 (6.3)	1,098	10.71 (3.4)
	n	%	*n*	%
Sex				
Female	372	67.8	521	47.4
Male	158	28.8	569	51.8
Ethnicity				
White—European	332	73.1		
White—North American	56	12.3		
Asian—South	17	3.7		
Asian—South East	8	1.8		
Black—African	10	2.2		
Black—North American	5	1.1		
Mixed	12	2.6		
Other	13	2.8		
Declined	95	17.3		
Employment				
Full-Time	276	52.2		
Part-Time	129	24.4		
Unpaid Work[a]	93	17.6		
Unemployed	14	2.6		
Current monthly household income				
< $2,000 USD	108	19.7		
$2,000-$3,999 USD	197	35.9		
$4,000-$5,999 USD	124	22.6		
$6,000-$7,999 USD	53	9.7		
$8,000-$9,999 USD	20	3.6		
$10,000-$11,999 USD	17	3.1		
$12,000-$13,999 USD	9	1.6		
≥ $14,000 USD	20	3.6		
Household annual income (2019)				
< $15,000 USD	21	3.8		
$15,000–$24,999 USD	49	8.9		
$25,000–$49,999 USD	154	28.1		
$50,000–$74,999 USD	133	24.3		
$75,000–$99,999 USD	90	16.4		
$100,000–$124,999 USD	48	8.8		
$125,000–$149,999 USD	17	3.1		
$150,000–$174,999 USD	19	3.5		
≥ $175,000 USD	17	3.1		
Relationship status				
Lone parent	44	8.0		
Living apart couple	8	1.5		
Married/common-law	495	90.2		
Same-sex partners	2	0.4		

[a] Includes homemaker, retired, disabled.

Measures

COVID-19 Disruption

Stressors related to the pandemic at T1 were assessed using a newly designed and validated measure (Prime et al., 2021). Twenty-five preliminary items were generated that broadly covered disruption across numerous aspects of life, including finances, basic needs, personal and family welfare, family relationships, career/education, and household responsibilities. Participants rated how applicable each stressor was to their households on a three-point scale: "*Not True*" (1), "*Somewhat True*" (2) and "*Very True*" (3). Given that the present study was concerned with modeling distal COVID-19 disruption on proximal family relationships and mental health, seven items pertaining to direct effects of the pandemic on family relationships were removed. This was a necessary step in order to guard against inflated correlations between the COVID-19 scale and outcome variables through the inclusion of shared items across scales. An additional nine items were removed that had low endorsement, were redundant or did not converge well with the scale. Final items included in the present study were concerned with the effects of the pandemic on income/economics (5 items: *decrease in income, financial debt, job loss, required government assistance, difficulty providing for family*), and pandemic-specific chaos (4 items: *could not access supplies, overwhelmed by news, crowded shopping centers, anxiety about danger to self and family*). These items were combined by taking a sum, which demonstrated good internal consistency (Cronbach's α =.75).

Caregiver Psychological Distress

Caregiver psychological distress was measured using the Kessler Psychological Distress Scale (K10; Kessler et al. 2002). The K10 is a widely utilized 10-item questionnaire that yields a global measure of distress related to depression and anxiety symptoms. Items assess a variety of feelings experienced in the past 30 days, with response options ranging from "*None of the time*" (1) to "*All of the time*" (5). Summed scores > 20 suggest the presence of clinical impairment (Andrews & Slade, 2001), though the scale can also be used continuously, as was done in the present study. Internal consistency in the current sample was excellent (Cronbach's α = .93).

Family Dysfunction

Family dysfunction was assessed using the six-item version of the General Functioning Subscale of the McMaster Family Assessment Device (FAD-GF6+; Boterhoven de Haan et al., 2015; Epstein et al., 1983). The scale assesses aspects of family life including planning, support, and cohesion. Total scores for the present study were calculated by averaging items which ranged from "*strongly agree*" (1) to "*strongly disagree*" (4). Averaged scores > 2 indicate problematic functioning. Large scale testing of the FAD-GF and FAD-GF6+ indicate strong reliability and validity (Boterhoven de Haan et al., 2015; Byles et al., 1988). Internal consistency in the current sample was good (Cronbach's α = .87).

Parenting Quality

Parenting quality was assessed using the 10-item parenting practices scale from the 2014 Ontario Child Health Study, an epidemiological study of Ontarian children aged 4 to 17 and their parents (Statistics Canada, 2017). Five items from this self-report questionnaire assessed the frequency of negative, coercive, and ineffective parenting and discipline practices, while five items measured positive caregiving, support, and enjoyment. Items were rated on a five-point scale ranging from "*Never*" (1) to "*Always*" (5). Negatively worded responses were reverse-coded and a composite was created by taking the sum across all items, where higher scores indicated higher quality parenting (more positivity, less negativity). Internal consistency in the current sample was good (older child: Cronbach's α =.83; younger child: Cronbach's α =.81).

Child Mental Health Problems

Children's mental health was measured using parent proxy reports on the Patient Reported Outcomes Measurement Information

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System (PROMIS) in the following domains: anger (5-item), anxiety (8-item), and depressive symptoms (6-item; PROMIS Health Organization, 2020). The PROMIS Measures are widely employed, robust instruments, developed using rigorous psychometric methodology (Health Measures, 2020; Irwin et al. 2012). Parents reported the frequency of difficulties on a scale ranging from "*Never*" (1) to "*Almost Always*" (5). Items were then summed to compute domain-specific scores and a total score (summation of all items), with higher scores indicating greater mental health problems. Internal consistency for the total scale was good (younger child: Cronbach's α = .94; older child: Cronbach's α = .95).

Covariates

All between-family paths controlled for 2019 (annual) household income, caregiver age and sex, sibling age gap (measured as a sibling standard deviation of age in years), and sibship sex composition (dummy variables for all male, and all female, with a mixed sex reference category.) Within-family paths were adjusted for child age and sex.

Analysis

The hypothesized model was evaluated using multilevel path analysis with random effects. Like many multilevel family studies (e.g., Browne et al., 2018, 2019), observations were structured so that children (level 1) were nested in families (level 2). Between-family effects (level 2) refer to associations among variables measured at the family level (i.e., there is one score or value for each family) and reflect differences across families or households (as in most single-child designs). Within-family effects (level 1) describe associations among variables that vary within families (i.e., there are different values for each child in the family), reflecting differences across siblings when group-mean (family) centering is used, as in the current study. Thus, the within-family component of this model is commonly described as a *sibling comparison model* (D'Onofrio et al. 2013). The model was fit using the Bayes estimator in Mplus 8 (Muthén & Muthén, 1998–2017).

Three components of the model permitted testing of study hypotheses. First, COVID-19 disruption was modeled as a predictor of family-level variables at time 1 (May 2020) and time 2 (July 2020), including caregiver distress, family dysfunction, family-average parenting quality, and family-average child mental health problems. Autocorrelations for family-level variables at time 1 and time 2 were included, in addition to all possible within-time covariances and cross-lag effects. These components tested the hypothesis that COVID-19 disruption was associated with various aspects of family life at the between-family level in a spillover fashion (H1). Second, family-level standard deviations for parenting quality and child mental health problems were included in the between-family component of the model, allowing us to test the hypothesis that siblings will be more different from one another in families where COVID-19 disruption is high (H2). Third, within families, bidirectional effects linking children's mental health and parenting quality were evaluated using a standard cross-lag configuration, where predictors are group-mean (family) centered. Within-family cross-lags were permitted to vary randomly across families and were predicted by time 1 between-family variables, allowing the examination of child-specific effects from children to parents and vice versa, which were expected to be more pronounced under settings of increased family stress (H3).

Model building took place in three steps. First, a model without indirect and random effects was estimated. This model permitted the evaluation of model fit using Bayesian Posterior Predictive Checking using Chi Square, which provides a 95% confidence interval of the difference in the observed and replicated chi square values. A good fitting model should have a 95% confidence interval that includes zero and a Posterior Predictive P-value (PPP) > .05. Next, indirect effects from COVID-19 disruption to time 2 variables via time 1 variables (at the between-family level) were added when there were significant cross lags linking time 1 predictors to different time 2 outcomes. These effects were evaluated using the Delta Method (i.e., computed as new model parameters using the model Constraint function), as bootstrapping indirect effects via the model Indirect function is unavailable within the multilevel Bayes procedure in Mplus 8. Thus, adding every possible indirect effect would significantly overload the model, which was already computationally demanding. We report unstandardized indirect effects. Finally, random effects for the within-family cross-lags were added along with time 1 between-family predictors of the random slopes (i.e., which are analogous to cross level interactions). Note, it is impossible to compute the PPP for the random effects model, since this would be akin to comparing the estimated model to itself. Deviance (DIC) is reported for both models. In Mplus 8, estimates from Bayesian estimation are reported with one-tailed p-values ("significant" results where $p < .025$) and 95% credibility intervals (see Asparouhov & Muthén, 2010). Our sample size is adequately powered for estimating unbiased variance components (Maas & Hox, 2005) and cross-level interactions (Scherbaum & Ferreter, 2009) in the context of medium-to-large effect sizes.

Missing Data

In terms of missing data, 514 (93.6%) of the original N = 549 families included at baseline were retained at two-month follow-up. Missing data were handled with the Bayes procedure in Mplus 8, which utilizes Monte Carlo Markov Chains (MCMC) based on the Gibbs sampler (Asparouhov & Muthén, 2010). This is analogous to Full Information Maximum Likelihood (FIML) estimation in traditional structural equation modeling and is based upon the missing at random (MAR) assumption.

Results

Descriptive statistics are presented in Table 2 and correlations are presented in Table 3. The initial model (no indirect or random effects) fit the data well, based upon the 95% confidence interval of the difference between the observed and replicated chi-square values (95% CI [−8.65, 94.65]) and PPP = .08. The DIC was 57881.91. Next, indirect and random effects were added (DIC = 54740.49). Figure 1 graphically presents the multilevel path model. Between-family components are in the upper pane, and within-family components are below. For ease of interpretation and to reduce clutter, within-time covariances at the between-family level are omitted and only significant parameters (those with 95% credibility intervals that do not include zero) are depicted. Complete model output is presented in the online supplemental material.

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Table 2
Descriptive Statistics of Measures

Measure	Time 1/May 2020 (N = 549)				Time 2/July 2020 (N = 514)			
	N	M (SD)	Min	Max	N	M (SD)	Min	Max
COVID disruption	549	14.95 (3.66)	9	26		—	—	—
Caregiver distress	546	20.12 (7.93)	10	49	503	19.08 (7.78)	10	46
Family dysfunction	549	10.01 (2.89)	6	24	512	8.75 (2.82)	6	24
Parenting quality								
Younger child	547	40.40 (5.07)	25	50	506	40.97 (4.91)	26	50
Older child	547	40.39 (5.44)	26	50	504	41.35 (5.27)	17	50
Child MH problems								
Younger child	546	32.95 (11.34)	19	80	505	31.45 (10.84)	19	73
Older child	547	33.84 (12.40)	19	81	504	31.63 (12.16)	19	80

Note. MH = mental health.

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Hypothesis 1: Spillover of COVID-19 Across the Family System

The hypothesis of COVID-19 disruption corresponding to family difficulties in a spillover fashion was supported. Higher levels of COVID-19 disruption were associated with greater caregiver distress, family dysfunction, and family-average mental health problems for children at time 1. Interestingly, the association between COVID-19 disruption and family-average parenting quality just missed the threshold for statistical significance (−.063, 95% credibility interval = −.126, .000, one-tailed p = .025). Over time, there were significant cross-lags connecting domains of family life across the follow-up period. Greater time 1 family dysfunction was associated with greater time 2 caregiver distress and family-average children's mental health problems, controlling for previous levels of these variables. Greater time 1 parenting quality was associated with lower levels of family dysfunction, controlling for previous levels. Greater time 1 family-average children's mental health problems were associated with greater time 2 caregiver distress and family dysfunction, and lower parenting quality, controlling for previous levels. In sum, COVID-19 stressors correspond to poorer family functioning and child mental health at time 1, and these factors "spill over" to predict caregiver distress, poorer parenting quality, and poorer mental health of children two months later.

Indirect effects for the implied pathways (i.e., significant links from COVID-19 disruption to one area of family life at time 1, to another area of family life at time 2) were examined through the definition of new model parameters. The indirect effect linking COVID-19 disruption to mental health via family dysfunction was significant for caregivers (B = .017, 95% Credibility Interval [.001, .005], one-tailed p = .018), and marginal for children, on average, (B = .017, [−.001, .051], one-tailed p = .032). COVID-19 disruption was also associated with greater child mental health problems, on average, which was associated with greater caregiver distress (B = .073, [.024, .132], one-tailed p = .001], and lower parenting quality (B = −.073, [−.114, −.042], one-tailed p < .001). Thus, spillover is simultaneously occurring from children-to-families and from families-to-caregivers.

Hypothesis 2: COVID-19 Corresponds to Greater Sibling Differences

The hypothesis for greater sibling differences under settings of pandemic stress was supported. Higher levels of COVID-19 disruption were associated with a larger family variance (SD) in children's mental health problems. In other words, siblings are more different in their mental health in families where COVID-19 disruption is high. Sibling differences in mental health problems persisted over time (i.e., significant, positive autocorrelation). Furthermore, family-average child mental health problems at time 1 predicted sibling differences in child mental health problems at time 2, controlling for previous levels. Similarly, lower parenting quality at time 1 predicted greater sibling differences in parenting quality at time 2, controlling for previous levels.

In terms of indirect effects, COVID-19 disruption corresponded to greater sibling differences in child mental health at time 2 via greater family-average child mental health problems at time 1 (B = .084, [.039, .143], one-tailed p < .001). This suggests that, in families where children are reported as having more difficulties, on average (i.e., higher family-average mental health problems), the effects are being driven by one sibling who is doing worse (i.e., siblings are more different from one another). This pattern grew over the two-month follow-up, suggesting a relative deterioration for the child who was initially doing worse.

Hypothesis 3: Child-Specific Effects Are Observable Within-Families

Moving to the within-family component of the model (see Figure 1, lower panel), within-family (sibling) differences in mental health and parenting were examined. First, a significant and negative within-time association between child-specific parenting quality and child-specific mental health problems at time 1 indicated that children with more mental health problems experienced lower levels of parenting quality compared to his or her sibling. Second, there was a significant autocorrelation for children's mental health problems and parenting, indicating that the sibling struggling with mental health at time 1 is more likely to be struggling at time 2 relative to his or her sibling, and the child receiving higher quality parenting at time 1 was also receiving this at time 2. Third, the cross-lag linking child mental health problems at time 1 and parenting quality at time 2 was inverse and statistically significant. The obverse direction was not significant, suggesting that the child with more mental health difficulties (compared to his or her sibling) received lower quality parenting over time, but that lower parenting quality was unrelated to subsequent child mental health problems.

Both of these parameters were permitted to vary randomly across families. In both cases, there was significant variance associated

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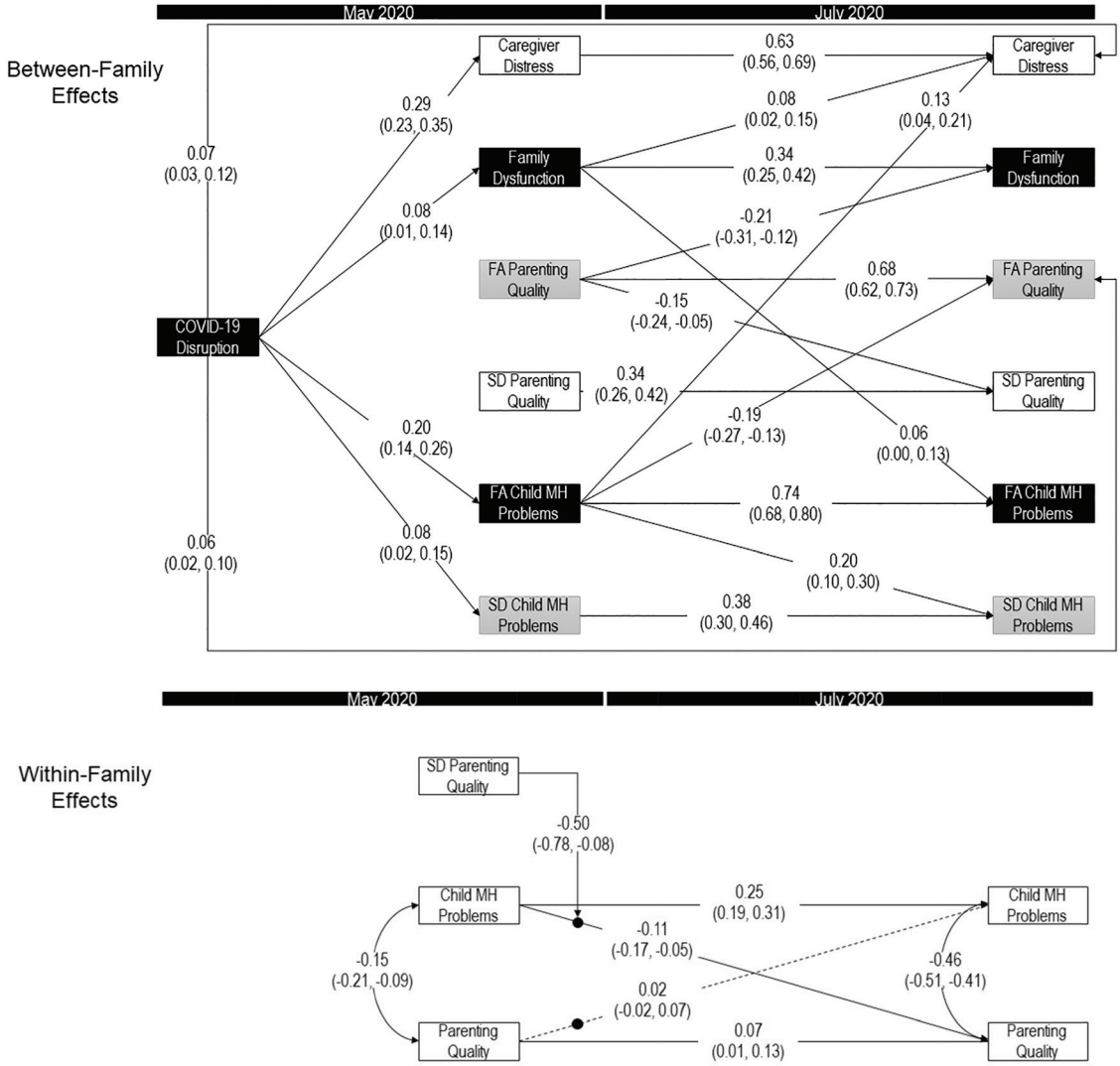
Table 3
Correlations Among Study Variables Between-Families (Above) and Within-Families (Below)

	Between family variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	16
1	COVID-19 disruption															
2	Caregiver distress T1	.42**														
3	Caregiver distress T2	.42**	.75**													
4	Family dysfunction T1	.10*	.26**	.29**												
5	Family dysfunction T2	.07	.12**	.17**	.40**											
6	FA parenting quality T1	−.06	−.26**	−.24**	−.35**	−.33**										
7	FA parenting quality T2	−.02	−.24**	−.32**	−.29**	−.39**	.78**									
8	*SD* parenting quality T1	−.03	.09*	.05	.12**	.11**	−.22**	−.16**								
9	*SD* parenting quality T2	.05	.08	.14**	.13**	.20**	−.24**	−.29**	.38**							
10	FA child MH problems T1	.26**	.48**	.46**	.26**	.24**	−.48**	−.50**	.11*	.15**						
11	FA child MH problems T2	.20**	.39**	.54**	.25**	.29**	−.39**	−.50**	.07	.14**	.80**					
12	*SD* child MH problems T1	.12**	.19**	.20**	.14**	.15**	−.15**	−.23**	.11**	.12**	.39**	.37**				
13	*SD* child MH problems T2	.12**	.17**	.24**	.08	.17**	−.18**	−.25**	.08	.18**	.35**	.45**	.46**			
14	Income 2019	−.25**	−.11**	−.12**	−.05	−.03	.00	.01	−.07	−.06	.00	.01	−.04	−0.05		
15	Caregiver age	−.12**	−.16**	−.13**	.05	.10*	−.03	−.02	.02	−.01	.01	.03	.04	.08	.21**	
16	Caregiver male	−.20**	−.17**	−.15**	−.09*	.04	−.09*	−.10*	.04	−.04	−.02	.00	−.01	−.02	.14**	.17**
	Within family variables	**A**	**B**	**C**	**D**	**E**	**F**									
A	Parenting quality T1															
B	Parenting quality T2	.71**														
C	Child MH problems T1	−.42**	−.43**													
D	Child MH problems T2	−.33**	−.44**	.74**												
E	Child age	.29	.02	−.03	−.03											
F	Child male	−.02	.03	−.03	−.08*	−.02										

Note. T1 = time 1 (May 2020); T2 = time 2 (July 2020); FA = family average; *SD* = standard deviation; MH = mental health.
* $p < .05$. ** $p < .01$.

Figure 1
Multilevel Family Stress Model Demonstrating Relationships Among Variables Between and Within-Families



Note. FA = family average; *SD* = standard deviation; MH = mental health.

with the random effects, indicating that households differ in the child-to-parent cross lag (σ^2 = .49, [.27, .76], one-tailed $p < .001$), as well as the parent-to-child cross lag (σ^2 = .49, [.20, .78], one-tailed $p < .001$). These random slopes were permitted to covary and the covariance was nonsignificant. The between-family variables at time 1 (including covariates) were treated as predictors of the random effects. The only significant predictor was the family variance (i.e., *SD*) in parenting quality predicting the effect of child-specific mental health on child-specific parenting quality. Specifically, the coefficient linking time 1 child-specific mental health problems to time 2 parenting quality was more negative among families who reported higher time 1 variance in parenting quality. Stated differently, when caregivers treat their children differently, their parenting is more strongly influenced by the mental health of particular children over time. In contrast, when parents treat their children relatively equally, the behavior of one particular child does not influence the parenting he or she receives. Associations among study variables with covariates are presented in the full model output (see online supplemental material).

Discussion

The present study highlights important associations between COVID-19 disruption, family relationships (i.e., general family functioning and parenting), and child/caregiver mental health. Longitudinal patterns are demonstrated between- and within-families, extending the literature on family relationships and mental health during the pandemic to include family-wide versus child-specific processes. We found that households struggle more when COVID-19 disruption is high, experiencing greater family dysfunction and more child mental health problems, which predict caregiver mental health and parenting behavior over time. Also,

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one particular child tends to have greater difficulties in a given family over time (in terms of mental health and quality of parenting received), suggesting that there is a pattern of intraindividual change that amplifies sibling differences. These findings provide a nuanced model connecting pandemic-related distress to child/caregiver mental health and family well-being, while also explicating the strength and direction of associations across the family unit.

COVID-19 and Spillover Across Family Systems

The associations linking pandemic stressors with family relationships and children's mental health are largely in accordance with the COVID-19 Family Disruption model outlined by Prime et al. (2020), in addition to decades of research articulating the developmental sequelae of financial hardship (Masarik & Conger, 2017), cumulative adversity (Repetti et al., 2002), and large-scale disasters (Masten, & Motti-Stefanidi, 2020). When COVID-19 disruption is high, caregivers are more distressed, families are more dysfunctional, and children have more mental health difficulties. Like other multilevel studies (Plamondon et al., 2018), we observed evidence of child-effects at the between-family level, suggesting that difficulties across the sibling subsystem (i.e., having multiple children with mental health concerns) can relate to lower quality parenting over time. Moreover, in line with a transactional framework (Sameroff, 2010), household COVID-19 disruption corresponded to greater mental health problems within the sibling subsystem, which was related to greater challenges in other areas of the family system (i.e., caregiver distress and parenting quality, on average).

The present study also demonstrated evidence for family-driven effects: COVID-19 disruption was associated with higher family dysfunction, which corresponded to more child and caregiver mental health difficulties (though the indirect effect for children was marginal). This pattern is consistent with ecological models of human development (Bronfenbrenner & Morris, 2007), as well as the traditional family stress model (Masarik & Conger, 2017), whereby the direction of influence flows from the distal ecology to the proximal family context, and onward to individual children and caregivers. In conjunction with the aforementioned child-effects, these family-effects highlight bidirectionality in spillover, while also suggesting that whole-family effects may be largely child versus caregiver driven. Indeed, failure to observe caregiver-driven pathways (i.e., effects linking earlier caregiver distress to later family relationships or child psychopathology) was contrary to hypotheses. Our study began two months into the pandemic, where increases in caregiver psychopathology were already reported (Cooke et al., 2020; Patrick et al. 2020). Thus, it is possible that caregiver-driven effects had already taken place prior to our investigation, and that we were capturing the resultant consequences of child psychopathology back on their caregivers and families. Additionally, cross-national contextual factors such as pandemic severity, school closure, and whether children required continuous supervision in the home may have also contributed to child challenges in the second follow-up. Longitudinal follow-up of this cohort and other samples will further uncover the ways in which COVID-19 stress emanates across domains of family life over time.

COVID-19 Disruption and Sibling Differences

It is important to consider a number of findings simultaneously to describe *how* and *why* siblings may be more different in mental health when pandemic stress is high. First, the association between COVID-19 disruption and caregiver distress suggest that parents' tolerance for challenging behavior may be reduced when pandemic stressors are high. Within the family stress paradigm, when parents are focused on supplying children with basic needs, they may not have the time, energy, or patience to respond optimally to children. Studies of differential parenting suggest that parents may especially lose patience with more difficult children (Jenkins et al., 2003; Plamondon et al. 2018), suggesting that caregiver resource dilution may play a role in the potentiation of sibling differences. Greater sibling differences under settings of shared family stress have also been explained through the notion of differential vulnerability and stress sensitization. One sibling, because of their own pattern of temperament and previous adaptation, is often more vulnerable to the effects of stress (Rende, 2012). In other words, the deterioration in the mental health of an individual child could recruit more negative parent–child interaction, corresponding to greater sibling differences across the whole family system. This pattern has been described in other sibling studies (Plamondon et al., 2018). Further, the present study suggests that child effects on parents may be greatest when parents are already struggling and, consequently, their levels of differential parenting are high.

In summary, COVID-19 disruption corresponds to greater mental health problems for particular children within the family, which relates to sibling differences in parenting quality, showcasing one way the pandemic "gets inside the family" to inform the relationships of certain parent–child dyads. Indeed, parents are most reactive and least positive to the child showing the highest levels of mental health difficulties, and their differential treatment of siblings persisted over a two-month period. These findings suggest that pandemic stressors are associated with intraindividual changes in child mental health, along with increased sibling differences, in terms of both parenting and psychological functioning.

Clinical Implications

The results of this study suggest possible family-centered foci for psychosocial interventions. While not directly evaluated, findings imply that prevention/intervention in one family subsystem could have positive ripple effects across the family unit. Models that support positive parenting and emotionally validating communication, while simultaneously considering whole-family dynamics, could be useful. This may be particularly true when considering the role of sibling differences in children's mental health and differential parenting during periods of family stress. Throughout COVID-19, when in-person work is less common, clinicians can engage with parents via telehealth to indirectly support the child who is experiencing the greatest difficulties, along with the sibling relationship. Simultaneously, interventions with a whole-family lens (e.g., family therapy) can extend this to supporting caregiver functioning and relationships, which will likely convey benefit to the family system.

Professionals have been encouraged to provide evidence-based psychotherapy to parents and children via virtual mediums to address mental health during and after COVID-19 (Racine, Birken, et al., 2020). Consideration of family-wide dynamics (which can be

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driven by individual children and/or the sibling subsystem) is likely important to promote the generalization of these strategies. There is a pressing need to evaluate the effectiveness of whole-family telehealth interventions. Following a more traditional family systems approach, family-directed telehealth services should allow for the involvement of multiple members of the family in treatment, with the goal of identifying and intervening on strained relationships (Carr, 2015). Clinical guidelines for "relational" telehealth therapies are growing, though empirical evaluations are needed (Burgoyne & Cohn, 2020). By using these findings to target specific subsystems and families who are vulnerable to the consequences of the pandemic, programming and policy can help families avoid further spillover of COVID-19 disruption into the family unit.

Limitations and Future Directions

The current study did not have a prepandemic assessment and, as a result, we cannot ascribe variation in outcomes as arising from the pandemic specifically. Second, this study is within-informant (i.e., the parent/caregiver). Thus, problems of shared method bias should be considered. Third, there were only two data collection points. A minimum of three time points are required to robustly examine transactional effects. Fourth, COVID-19 disruption was only assessed at the first wave. Therefore, pandemic disruption is considered constant even though this is likely to fluctuate over time. Similarly, there are other contextual factors that were not modeled, such as time of year and presence in the classroom (e.g., most children should have been in school or online-learning for the first assessment, but on summer holiday for the second). Fifth, there are processes outlined in the COVID-19 Family Disruption framework that were not included in the current study, such as family/individual strengths and other protective factors. Sixth, our study was powered to detect medium-to-large effects in terms of cross-level interactions (i.e., family-level predictors of random slopes), meaning we may be underpowered to detect very small to small effects. Seventh, while our sample includes multiple countries, we did not specifically take national differences into account due to marked differences in cell sizes. It is possible that associations among some variables (e.g., COVID-19 disruption and outcomes) are partially attributable to country differences in pandemic severity. Lastly, while implications for services emerge from this study, we did not consider the services that families are naturally receiving. International, clinically informative research during the pandemic will certainly be instrumental in supporting families during this stressful period.

Conclusion

The current study utilized a longitudinal sibling comparison design to demonstrate family-wide and child-specific associations with pandemic-related disruption. Findings were aligned with family spillover and transactional hypotheses, consistent with recent theoretical formulations of the way in which higher levels of pandemic disruption relate to more family dysfunction and child mental health problems. We show that the mental health of one child in particular may be more strongly tied to pandemic disruption, and that this, in turn, is related to quality of parenting received by children over time. Ultimately, these patterns may correspond to further decline in family functioning, though paradigms more amenable to causal inference and direction of effects are warranted, such as longitudinal cohorts that began before the pandemic. Nevertheless, our exploratory study underscores the importance of family-based services that address the needs of whole families during and after the pandemic.

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